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Heatwaves, Rooftop Solar and Data Centres Force Rethink of India's Power Sector Planning

ndia is no longer facing a conventional power supply challenge, as heatwaves, cooling demand, ooftop solar, storage needs and data centres change when and how electricity is produced and onsumed.

Puja Das

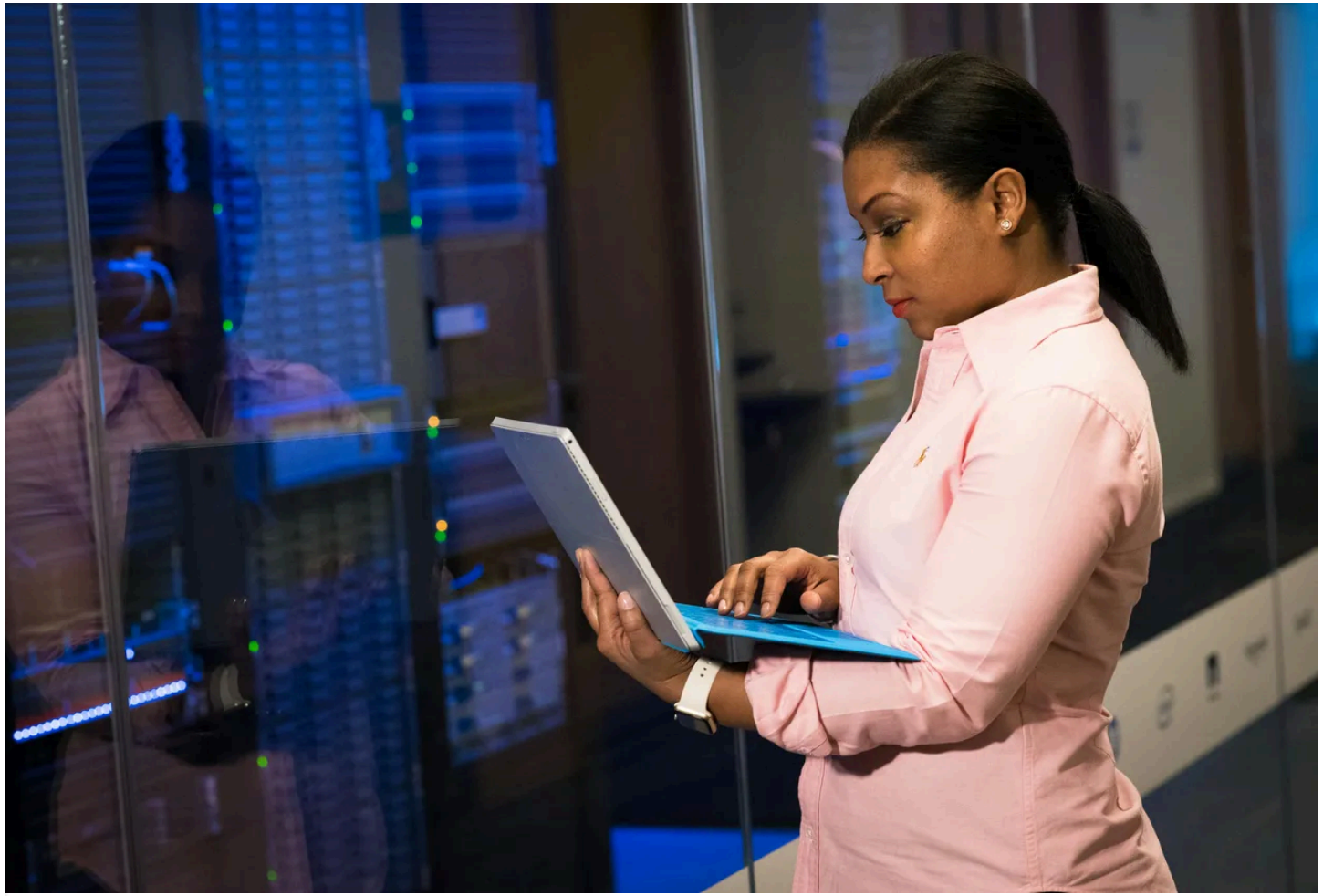
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ıdia’s power sector is entering one of its most complex transition phases as rising heat :ress, rapid renewable energy expansion, cooling demand, decentralised renewable ystems (DRE) and emerging industries such as data centres reshape electricity lanning.

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t a policy discussion on May 13, 2026 in Delhi on India's energy transition and grid reparedness, government officials, renewable energy developers, market experts and esearchers said India was no longer dealing with a conventional supply challenge. lectricity demand patterns, generation sources and transmission needs are all hanging at the same time, with heat emerging as a major stress factor.

ooling devices such as fans, air conditioners and air coolers account for 40 per cent f household electricity use, according to a new survey by energy efficiency non-profit LASP and the Bureau of Energy Efficiency. Provisional data from the Central Electricity uthority (CEA), showed the residential sector accounted for 25 per cent of India's total lectricity consumption in 2024-25, growing annually at 6 per cent.

xperts at the discussion warned that prolonged and unpredictable heatwaves are creasingly affecting transmission and distribution losses, thermal plant efficiency, coal gistics, voltage stability and evening peak demand.

On May 30, 2024 and again on April 25, 2026, the demand suddenly started increasing the afternoon and by around 2.30 pm to 3.30 pm we reached the highest peak," said ijay Menghani, chief engineer for emerging technologies and innovation at the CEA, at he Energy Dialogue of the India Heat Summit organised by research-based consulting rm Climate Trends.

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ndia touched a peak demand of 256 GW during the heatwave period, with demand sing by nearly 20 GW within two hours. Menghani said the grid had remained resilient ecause of planning by central agencies, state governments and developers and added hat the country is also increasing air-conditioning capacity by 20 per cent to 25 per ent annually while preparing for additional demand from data centres and anufacturing.

*W*e have planned up to 2047. Every additional increment in transmission will bring elief," he said, adding that transmission approvals that earlier took nearly three years re now being cleared within three months after reforms introduced in August 2022.

ooftop solar pressures

ndia is expanding renewable energy capacity while preparing for deeper electrification f transport, cooking, manufacturing and digital infrastructure. But the rapid growth of ecentralised renewable energy, especially rooftop solar, is creating fresh operational hallenges even as it helps reduce transmission and distribution losses.

*W*hen we increase DRE like rooftop solar, both the availability of location as well as eneration is not quite available to the grid operator," Menghani said, warning that udden cloud cover could reduce large amounts of rooftop generation at the same time.

nother issue is voltage instability caused by sudden spikes in rooftop solar injection at ıe local distribution level. Menghani told *Down To Earth* that the government would troduce new inverter technology within a year, built into the solar rooftop package.

round 10,000MW of rooftop capacity operating during daytime hours has already educed transmission and distribution losses by nearly 2 per cent annually, he added.

ishal Pandya, managing director and co-founder of energy platform REConnect nergy, said climate change and renewable energy variability were also altering ectricity markets.

Nifty 50 volatility ranges between 19 and 22 per cent. Bitcoin is around 80 per cent nnualised volatility, while electricity in 2024 touched nearly 300 per cent," he said.

ccording to him, climate-driven demand spikes and fluctuating renewable output are ıaking electricity operations significantly more complex. He said humidity-driven "feels ke" temperatures would increasingly affect cooling demand.

enewable energy already accounts for nearly 50 per cent of installed capacity, while ecentralised systems in states such as Rajasthan and Gujarat meet 20 per cent to 30 er cent of afternoon demand.

n solar hours, the average market price would be Rs 1.5 per unit. Night hours touch Rs
) per unit, which is the cap," Pandya said, warning that evening supply shortages
ould trigger sharp price spikes.

Evening peak challenge

aloni Sachdeva Michael, energy specialist at the Institute for Energy Economics and
inancial Analysis, or IEEFA, said India's challenge was no longer only about meeting
igher demand, but also managing when that demand occurs.

iting findings from the State Electricity Transition report prepared jointly by IEEFA and
mber, she said India recorded peak demand of 256GW on April 25, 2026, around 9 per
ent higher than the corresponding period last year.

Delhi met its peak demand at 7GW in the month of March. Earlier, Delhi's peak demand
sed to come in June and July," Michael said.

ne third of the April 2026 peak demand was met by renewables, with solar alone
ontributing around 57GW. But evening demand remains the biggest challenge as solar
eneration declines and night-time temperatures continue to rise.

Night-time temperatures are also rising, which peaks the cooling demand," she said.

lichael stressed the need for large-scale battery energy storage systems and pumped ydro projects. India currently has around 6GW of battery energy storage deployment gainst a target of 61GW by 2030.

ndia's total solar potential stood at around 3,000 GW as of March 2025, but only 4.5 per ent had been utilised by March 2026, said Michael. Rajasthan has used around 8 per ent of its renewable energy potential, Maharashtra around 9 per cent and Gujarat bout 15 per cent.

ut rising renewable deployment has also intensified curtailment, where available enewable electricity cannot be evacuated because of inadequate transmission or sufficient demand.

ubrahmanyam Pulipaka, chief executive officer of the National Solar Energy Federation f India, said curtailment had two dimensions: low-demand curtailment and evacuation-elated curtailment, where projects are ready before transmission infrastructure. "The nancial implication is higher in the latter scenario," he said.

ndia has around 500,000 km of transmission lines, enough to technically support more an 300 GW of renewable energy, but transmission systems are struggling to keep ace with generation projects. Delays are linked to right-of-way disputes, wildlife earances and equipment shortages.

ulipaka said India's electricity system was moving towards co-located renewable nergy and storage systems that use the same transmission infrastructure more fficiently. "Energy storage is going to be the key," he said.

lew demand centres

ebajit Palit, centre head of the Centre for Climate Change and Energy Transition at the hintan Research Foundation, said India's bigger challenge was insufficient ectrification of overall energy demand, rather than a lack of renewable generation.

ndia consumes around 1,700 billion units of electricity annually, representing only about 2 per cent of total energy demand.

t should have been maybe 3,000 BU by now and all the problems of solar would be olved," he said.

oal currently contributes roughly 1,200BU annually and accounts for nearly 70 per ent of grid generation. According to NITI Aayog projections cited by Palit, coal could till generate around 1,300 BU in 2047 even if its share falls to 34 per cent, because verall electricity demand will rise sharply.

Renewables are not displacing coal. They are spacing coal," he said.

alit also raised concerns over the rapid expansion of data centres and their nplications for electricity demand and urban heat. Estimates by central think tank NITI ayog suggest data centres could consume around 45 GW of electricity by 2045, early equivalent to India’s current hydropower capacity.

tudies by Stanford and Cambridge universities also suggest data centres can increase mperatures by 2 per cent to 9 per cent within a 10 km radius.

*W*hat would be the impact of that on the cooling load?” Palit asked, questioning hether high-heat states such as Rajasthan should aggressively promote data centre xpansion without factoring in secondary heat impacts.

tates lead transition

he State Electricity Transition report identified Karnataka, Haryana and Himachal radesh among the leading states on decarbonisation metrics because of renewable nergy deployment and hydroelectric capacity. Delhi and Haryana were cited as strong erformers in distributed solar deployment, while Bihar and Chhattisgarh showed nprovements in DISCOM performance and smart meter implementation.

elhi currently leads the country in electric vehicle penetration at around 11 per cent.

he report also highlighted growing use of solar-aligned time-of-day tariffs, green ariffs and energy efficiency interventions to manage demand growth.

lichael said decentralised renewable energy, smart metering and energy efficiency ould become critical tools for managing future cooling demand and reducing stress on ne electricity network.

espite operational challenges, officials, developers and researchers agreed that India's ectricity transition remains on track and that many current problems reflect the speed f transformation rather than structural failure.

he key challenge ahead will be synchronising generation growth, transmission xpansion, storage deployment and rising electrification demand.

We should not fear because of some small problems," Menghani said. "When we tarted this energy transition journey, we chose a path."

his story was first published on Down to Earth.

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